
Cities Under Pressure: A Quantitative Framework

Summary

Urban resilience and sustainability have become critical challenges as cities worldwide confront increasing climate extremes, public health crises, and economic instability. However, existing approaches often rely on fragmented indicators or region-specific standards, limiting their usefulness for global benchmarking and long-term planning. To address these limitations, this study proposes an integrated **Urban Resilience and Sustainability Index (URSI)** that combines multi-source socio-economic and environmental data within a unified quantitative framework.

The URSI framework captures the multidimensional nature of urban resilience through a structured indicator system spanning **four key dimensions**. These dimensions are represented by **15 quantitative indicators** designed to reflect both short-term shock resistance and long-term adaptive capacity. Infrastructure indicators describe the reliability of essential urban systems, resource–environment indicators measure exposure to environmental stressors.

To determine indicator importance while minimizing subjectivity, a **hybrid weighting scheme** combining the **Analytic Network Process (ANP)** and the **CRITIC method** is employed. ANP incorporates expert knowledge and interdependencies among indicators, while CRITIC assigns objective weights based on data variability and information content. The coupled weights are synthesized using the **Rank-Sum Ratio (RSR)** method.

The results reveal a clear stratification of urban resilience and sustainability. Cities in developed economies generally achieve higher URSI values, reflecting stronger infrastructure systems, greater healthcare capacity, and more stable economic conditions. In contrast, cities from developing regions exhibit lower average URSI scores and greater dispersion, indicating higher vulnerability and uneven development across dimensions. Specifically, top-performing cities typically score above **0.75**, whereas lower-ranked cities remain below **0.60**, highlighting disparities in healthcare provision, environmental pressure, and economic stability.

Beyond static evaluation, the URSI framework is extended to a dynamic context using a **Long Short-Term Memory (LSTM)**–based prediction model. Leveraging the temporal continuity and nonlinear evolution of URSI, the model forecasts urban resilience trends over **2025–2029**, achieving an out-of-sample performance of $R^2 = 0.689$. The projections indicate that urban resilience evolves gradually, with **mid-ranked cities showing the greatest potential for improvement** under targeted policy interventions, while lower-ranked cities face slower progress due to structural constraints.

Overall, the proposed URSI framework provides a **robust and forward-looking tool** for benchmarking and forecasting urban resilience, offering actionable insights for sustainable urban planning in an increasingly uncertain global environment.

Keywords: Urban resilience; Sustainable development; Composite index; Multi-criteria evaluation; Time-series prediction

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1 Introduction

1.1 Problem Background

In the context of increasing climate change and public emergencies, “urban resilience”—a city’s capacity to recover, adapt and sustain development—has become a core focus in planning [1]. Meanwhile, progress toward the Sustainable Development Goals (SDGs) requires cities to balance economic growth, social equity and environmental protection [2]. This highlights the need for a comprehensive “**Urban Resilience and Sustainability Index (URSI)**.”

The index integrates factors such as infrastructure, governance, equity and ecology [3] to assess resilience and support strategic planning. Its application helps cities identify weaknesses, allocate resources efficiently and improve risk response. Ultimately, it can reduce disaster impacts, promote low-carbon transition and contribute to global sustainable development.

Global Trends of Public Health and Extreme Climate Events (2010-2025)

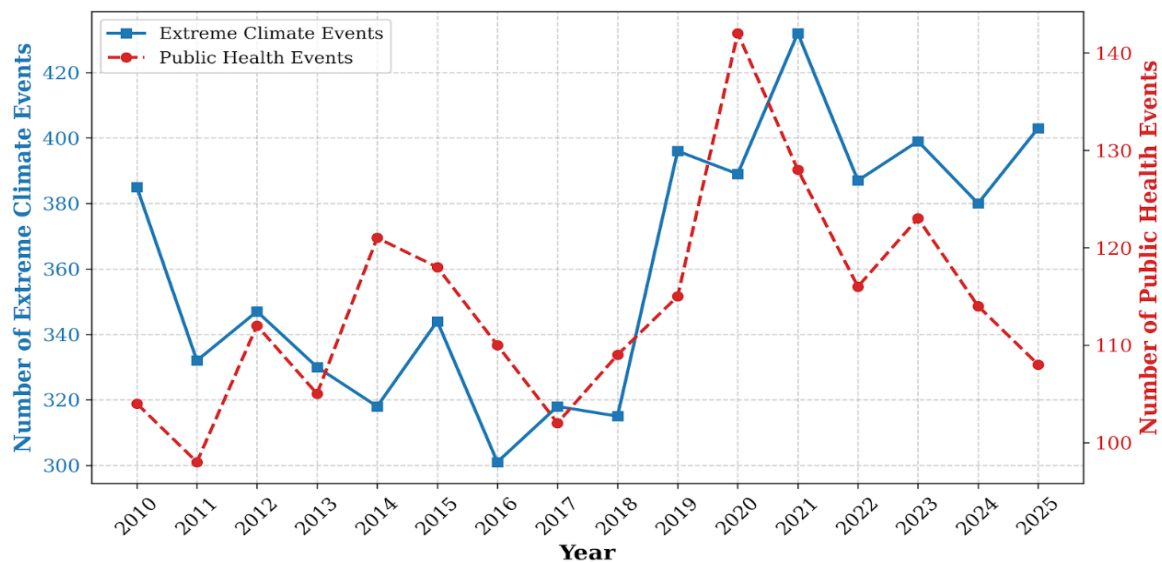


Figure 1. Temporal trends of global major public health emergencies and extreme climate events (2010-2025).

1.2 Restatement of the Problem

Considering the background information and restricted conditions identified in the problem statement, we need to solve the following problems:

- Build a scientific system of sustainable and resilient cities, and clarify the specific meaning of indicators at all levels and the calibre of data statistics.
- Select representative cities and collect relevant data.
- On this basis, build a mathematical model for the evaluation of resilience and sustainability, and realize the quantitative analysis of urban resilience and sustainable development.
- Predict the trend of URSI in the next five years.
- Write no more than one page of non-technical reports.

1.3 Our Work

To address the problem of quantitatively evaluating and forecasting urban resilience and sustainability, we developed a systematic modeling framework consisting of four main stages.

- First, we constructed a comprehensive URSI indicator system by integrating infrastructure reliability, environmental carrying capacity, economic resilience, and social governance capacity. Publicly available statistical data and satellite-derived indicators were collected for 20 representative cities worldwide, ensuring long-term consistency and cross-city comparability.
- Second, to determine indicator importance, we proposed a coupled ANP–CRITIC weighting method. ANP captures the interdependence among indicators from a structural perspective, while CRITIC reflects information content and variability in the data. The two sets of weights were integrated to reduce subjectivity and improve robustness.
- Third, the Rank-Sum Ratio (RSR) model was employed to synthesize weighted indicators into a single URSI score, allowing stable ranking and classification of cities under heterogeneous indicator scales. Cluster analysis was further conducted to identify cities with similar resilience profiles.
- Finally, we developed an LSTM-based prediction model to forecast future URSI trends. By learning temporal dependencies from historical URSI trajectories, the model provides medium-term projections and supports forward-looking resilience planning.

The proposed framework enables both comparative assessment and dynamic prediction of urban resilience, offering practical insights for policymakers and urban planners (**Fig 2**).

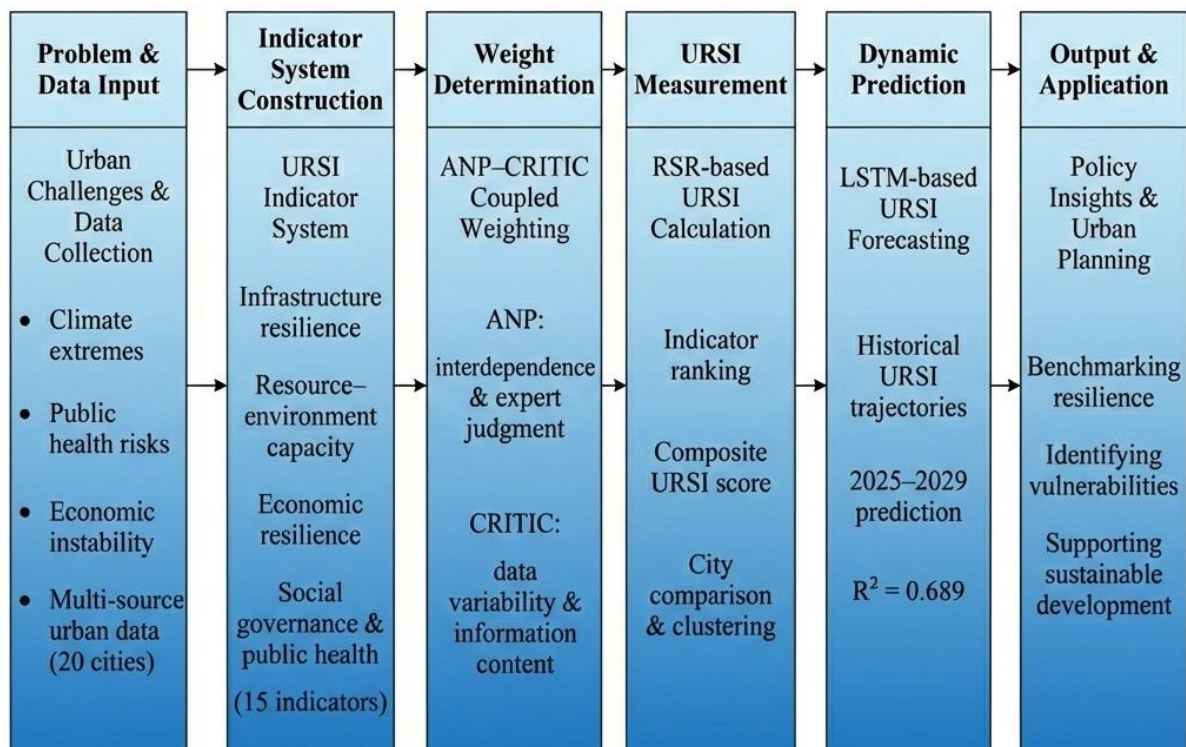


Figure 2. Study flowchart.

2 Assumptions

In order to solve the problems and more easily consider real-life situations, we make the following reasonable basic assumptions, each of which is properly justified.

- **No extreme disruptions.** Cities are assumed to be free from war or large-scale violence, as such events invalidate resilience and sustainability assessment.
- **Continuous socio-economic activity.** Urban production, consumption, and trade are assumed to persist; without economic activity, resilience is meaningless.
- **Stable population and ecological constraints.** Urban population size and development are assumed to remain stable and within ecological carrying capacity to ensure long-term sustainability.
- **Effective policy execution.** Urban development is assumed to comply with relevant policies and regulations, with plans implemented in practice[4].

3 Notations

The key mathematical notations used in this paper are listed in **Table 1**.

Table 1: Notations used in this paper

Symbol	Description
$y_i^{\max}$	The maximum values of the i-th indicator
$y_i^{\min}$	The minimum values of the i-th indicator
$[y_i^{\min}, y_i^{\max}]$	The actual range of the i-th indicator
$[a, b]$	The optimal interval
W_{pq}	represents the mean of the indicators in k schemes and σ_j
$\bar{X}$	Indicates the correlation coefficient between X_m and X_n
r_{mn}	Calculated coefficient
RSR_{max}	The maximum RSR value
RSR_{min}	The minimum RSR value
$\Phi^{-1}(\cdot)$	denotes the inverse standard normal distribution
$\hat{y}_{t+1}$	The predicted URSI value for time $t+1$
σ	Denotes the sigmoid function
$\odot$	Denotes element-wise multiplication

Other notations instructions will be given in the text.

4 Construction of the URSI Indicator System

4.1 Infrastructure & Engineering Resilience

- **Population Density (PopDensity) / persons·km⁻² – x_1**
Population density represents the intensity of human exposure to urban hazards and the de-gree of load imposed on infrastructure systems.
- **Built-up Area Ratio (BuiltRatio) / % – x_2**
The built-up area ratio quantifies the proportion of impervious surfaces within urban space, reflecting the rigidity and engineering dominance of land use patterns.
- **Road Network Density (RoadDensity) / km·km⁻² – x_3**
Road network density measures the spatial redundancy and accessibility of urban transportation infrastructure.
- **Electricity Access Reliability / index – x_4**
Annual Nighttime Light Stability Index or Power Outage Frequency Proxy
This indicator captures the reliability of urban energy infrastructure by reflecting the stability of electricity supply.

4.2 Resource & Environmental Carrying Capacity

- **Flood Exposure Index (FloodExp) / index – x_5**
Flood exposure index reflects the proportion of urban areas and population exposed to fluvial or pluvial flood hazards.
- **Nitrogen Dioxide Concentration (NO₂) / μg·m⁻³ – x_6**
This indicator serves as a high-fidelity proxy for atmospheric degradation resulting from urban mobility and industrial activities. It characterizes the baseline environmental stress thereby exerting a downward pressure on overall resilience.
- **Urban Greenness (NDVI) / index (0–1) – x_7**
This is a positive benefit indicator that encapsulates a city's innate ecosystem service capacity. NDVI functions as a critical parameter for nature-based buffering—facilitating micro-climate regulation and hydrologic attenuation
- **Land Surface Temperature (LST) / °C – x_8**
LST provides a direct spatial observation of thermal hazard intensity and the magnitude of the Urban Heat Island (UHI) effect. LST represents a detrimental factor to the physical integrity of urban infrastructure.

4.3 Economic & Financial Resilience

- **GDP per Capita (GDPpc) / USD/person⁻¹ – x_9**
GDP per capita delineates the fiscal ceiling for resilience-oriented investments. It reflects the economic elasticity available to subsidize adaptive infrastructure, directly contributing to a city's robust recovery potential.
- **Unemployment Rate (Unemployment) / % – x_{10}**

A high unemployment rate signifies diminished grassroots recovery capacity and increased household vulnerability; in the model, it functions as a limiting factor that impedes the systemic velocity of restoration following a crisis.

➤ **Nighttime Lights Growth (NTLGrowth) / % – x_{11}**

This indicator tracks annual changes in nighttime lights to monitor economic vitality. It serves as a proxy for regional development potential and investment attractiveness, which are key drivers of long-term resilience.

➤ **Nighttime Lights Volatility (NTLVolatility) / index – x_{12}**

This indicator measures multi-year fluctuations in nighttime light intensity (e.g., variability relative to the mean). It reflects economical stability and sensitivity to external shocks; higher volatility indicates more susceptibility to interference.

4.4 Social Governance & Public Health Resilience

➤ **Hospital Beds (Beds) / beds·1,000 persons⁻¹ – x_{13}**

This indicator reflects the number of hospital beds available per 1,000 population, measuring the healthcare system's surge capacity during pandemics, disasters, and mass-casualty events.

➤ **Physicians (Physicians) / physicians·1,000 persons⁻¹ – x_{14}**

This indicator represents the density of physicians per 1,000 population, reflecting the professional capacity for diagnosis, treatment, and sustained care during prolonged public health emergencies.

➤ **Life Expectancy (LifeExp) / years – x_{15}**

This indicator provides a comprehensive measure of population health. It captures baseline public health resilience and reveals underlying vulnerabilities when facing environmental and socio-economic shocks.

The URSI index system, as summarized above, is shown in **Figure 3**.

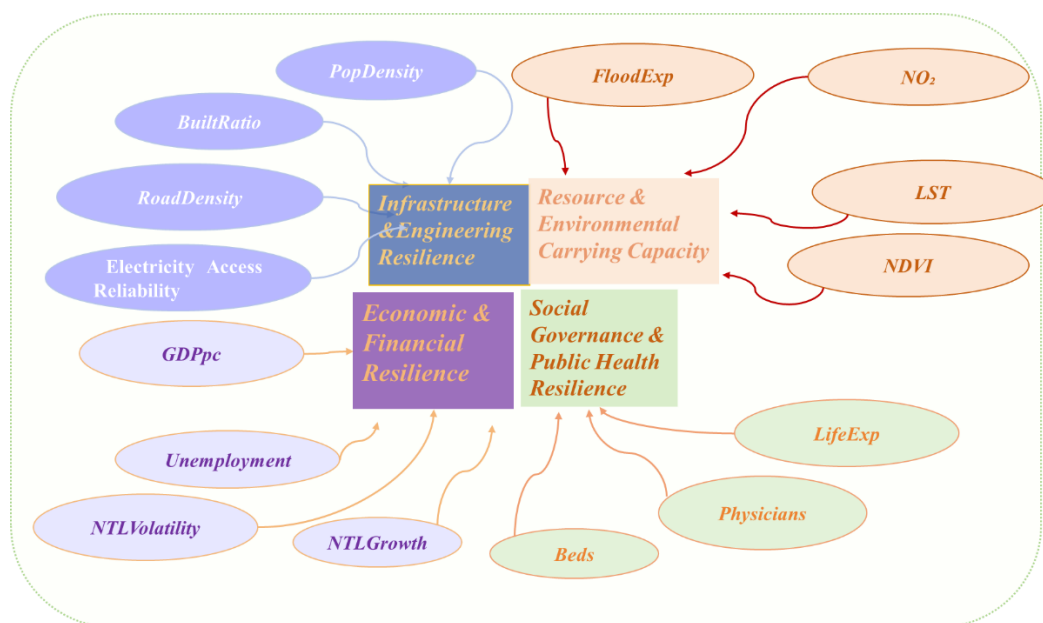


Figure 3 URSI Index System

5 The URSI city selection and data collection

5.1 The URSI city selection

In this study, we selected a total of 20 cities worldwide, comprising 10 from developing countries and 10 from developed economies. These include Mumbai, São Paulo, Mexico City, Jakarta, Cairo, Lagos, Bangkok, Buenos Aires, Manila, Nairobi from developing regions and Tokyo, New York, London, Paris, Berlin, Sydney, Toronto, Seoul, Zurich, Stockholm from developed regions. The results of the selected cities are shown in **Figure 4 [5]**.

The selection is based on the four core dimensions of the Urban Resilience and Sustainability Index (URSI), covering infrastructure, resource-environment, economy, and social governance. This allows for a structured comparison of resilience strategies across different developmental contexts, ensuring that the findings are both reliable and broadly applicable.



Figure 4 Selection of research city

5.2 Data Collection

To ensure transparency, reproducibility, and cross-city comparability, this study relies exclusively on widely used public data portals that provide long-term, well-documented urban, environmental, and socio-economic indicators. The selected sources cover four core dimensions of the Urban Resilience and Sustainability Index (URSI): infrastructure systems, resource and environmental conditions, economic performance, and public health and social capacity.

Each data source links to an official landing page offering downloadable time-series datasets or application programming interfaces (APIs). For remotely sensed indicators (e.g., PM2.5, NDVI, land surface temperature, nighttime lights, built-up area), globally consistent products were used to avoid inconsistencies arising from local statistical definitions. Socio-

economic and public health indicators were obtained from internationally recognized institutions to ensure comparability across developed and developing economies[6-7].

All datasets were checked for unit consistency, temporal coverage, and methodological revisions. When parallel datasets were available, the most recent and well-documented series was preferred. Time spans were harmonized across cities prior to normalization and index construction (**Table 2**).

Table 2. Data Collection for URSI Construction

Data source link	Example variables
https://ghsl.jrc.ec.europa.eu/	Population density, built-up area share, built-up expansion rate, urban spatial extent
https://eogdata.mines.edu/products/vnl/	Annual nighttime lights intensity, nighttime lights growth, nighttime lights volatility
https://www.who.int/data/gho/	PM2.5 concentration, NO ₂ concentration, urban air quality indicators
https://data.worldbank.org/	GDP per capita (US\$), unemployment rate, hospital beds per 1,000 people, physicians per 1,000 people, life expectancy
https://lpdaac.usgs.gov/	NDVI (MODIS), land surface temperature (LST), vegetation and thermal indicators
https://developers.google.com/earth-engine/	Remote sensing data access and city-level aggregation of NDVI, LST, and built-up surfaces

6 Measurement and Assessment of URSI

Prior to the measurement of **URSI**, the raw data are first preprocessed to ensure comparability across indicators. On this basis, the **Analytic Network Process (ANP)** and the **CRITIC** method are employed to derive subjective weights and objective weights, respectively. The two sets of weights are then integrated through the Lagrange multiplier method to obtain coupled weights. Finally, the coupled weights are incorporated into the **RSR model** to measure the URSI of each city[8].

6.1 Data Pre-processing

The necessary step in processing the data is to standardize different types of indicators to make them comparable and suitable for weighting.

Benefit-based Indicators

For benefit-based indicators, a larger value indicates a better condition. The normalization formula is given by:

$$x_{ji} = \frac{y_{ji} - y_i^{\min}}{y_i^{\max} - y_i^{\min}} \quad (1)$$

where $y_i^{\max}$ and $y_i^{\min}$ represent the maximum and minimum values of the i -th indicator, respectively.

Cost-based Indicators

For cost-based indicators, a smaller value is preferred. The normalization formula is:

$$x_{ji} = \frac{y_i^{\max} - y_{ji}}{y_i^{\max} - y_i^{\min}} \quad (2)$$

where $y_i^{\max}$ and $y_i^{\min}$ are defined as above.

Interval-based Indicators

For interval-based indicators, the optimal value lies within a specific interval $[a,b]$. The closer the indicator value is to this interval, the better. The normalization formula is defined as:

$$x_{ji} = \begin{cases} \frac{y_{ji} - y_i^{\min}}{a - y_i^{\min}}, & y_i^{\min} \leq y_{ji} < a \\ 1, & a \leq y_{ji} \leq b \\ \frac{y_i^{\max} - y_{ji}}{y_i^{\max} - b}, & b < y_{ji} \leq y_i^{\max} \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

Let $[y_i^{\min}, y_i^{\max}]$ be the actual range of the i -th indicator, and let $[a,b]$ be the optimal interval. This formula ensures that the normalized value reaches 1 within the optimal interval and decreases linearly outside it until it reaches 0 at the extremes.

6.2 URSI Index Weight Calculation

6.2.1 The ANP Method

ANP accounts for interdependencies and feedback among criteria by constructing a supermatrix and deriving weights from its limiting priorities.

Step1: Build pairwise comparison matrices

For each influence relationship, construct a judgment matrix:

$$A = [a_{uv}]_{n \times n} \quad a_{uv} > 0 \quad a_{uv} = \frac{1}{a_{vu}} \quad (4)$$

Step2: Compute local priorities and consistency

Obtain w by extracting the eigenvector associated with the largest eigenvalue:

$$Aw = \lambda_{\max} w \quad \sum_1^n w_u = 1 \quad (5)$$

Consistency test:

$$CI = \frac{\lambda_{\max} - n}{n - 1} \quad (6)$$

$$CR = \frac{CI}{RI} \quad (7)$$

Typically require $CR < 0.10$.

Step3: Construct the unweighted super matrix

Assemble all local priorities into the super matrix:

$$W = \begin{pmatrix} W_{11} & \cdots & W_{1g} \\ \vdots & \ddots & \vdots \\ W_{g1} & \cdots & W_{gg} \end{pmatrix} \quad (8)$$

where W_{pq} represents the influence of cluster G_q on G_p .

Step4: Weighted super matrix and final ANP weights

Let $\alpha = (\alpha_1, \dots, \alpha_g)^\top$ be cluster weights, $\sum_{q=1}^g \alpha_q = 1$ The weight super matrix is:

$$W^* = W \cdot \text{diag}(\alpha_1, \dots, \alpha_g) \quad (9)$$

Compute the limit super matrix:

$$W^{(\infty)} = \lim_{n \rightarrow \infty} (W^{(*)})^n \quad (10)$$

The final ANP weights are taken from any column of $W^{(\infty)}$

$$W_{ANP} = (W_1, \dots, W_N)^\top, \sum_{i=1}^N W_i = \mathbf{1} \quad (11)$$

6.2.2 The CRITIC Method

The CRITIC method is an objective weighting technique used in multi-criteria decision making (MCDM). It determines indicator weights by simultaneously considering the **contrast intensity** (the amount of information contained in each indicator) and the **conflict** (the correlation) between indicators. Indicators with greater variation and lower correlation with others are assigned higher weights [9].

Step 1: Calculate the Standard Deviation of Each Indicator

The standard deviation σ_j measures the contrast intensity or spread of values for the j -th indicator across all evaluated objects. A larger σ_j suggests the indicator has a greater power to discriminate between objects.

$$\sigma_j = \sqrt{\frac{1}{m-1} \sum_{i=1}^m (x_{ij} - \bar{x}_j)^2} \quad (12)$$

where: m is the number of evaluation objects (e.g., cities). x_{ij} is the normalized value of the j -th indicator for the i -th object. $\bar{x}_j$ is the mean value of the j -th indicator across all objects.

Step 2: Calculate the Correlation Coefficient Matrix

The correlation coefficient r_{jk} quantifies the linear relationship (conflict) between indicator j and indicator k . A high positive correlation implies redundancy in information.

$$r_{jk} = \frac{\sum_{i=1}^m (x_{ij} - \bar{x}_j)(x_{ik} - \bar{x}_k)}{\sqrt{\sum_{i=1}^m (x_{ij} - \bar{x}_j)^2} \sqrt{\sum_{i=1}^m (x_{ik} - \bar{x}_k)^2}} \quad (13)$$

The result is a symmetric correlation matrix $R = [r_{jk}]_{n \times n}$, where n is the number of indicators.

Step 3: Calculate the Amount of Contained Information

The total information content C_j for the j -th indicator combines its contrast intensity and its conflict with all other indicators.

$$C_j = \sigma_j \times \sum_{k=1}^n (1 - |r_{jk}|) \quad (14)$$

Here, $(1 - |r_{jk}|)$ represents the degree of conflict between indicators j and k . A higher C_j

value indicates that the indicator provides more independent and significant information.

Step 4: Determine the Objective Weights

The final objective weight $w_j^{(\text{CRITIC})}$ for the j -th indicator is obtained by normalizing its information content relative to the sum for all indicators.

$$w_j^{(\text{CRITIC})} = \frac{C_j}{\sum_{k=1}^n C_k} \quad (15)$$

These weights satisfy $\sum_{j=1}^n w_j^{(\text{CRITIC})} = 1$ and $w_j^{(\text{CRITIC})} \in [0,1]$.

6.2.3 ANP–CRITIC Coupling Weight Determination

To calculate the coupling weight of ANP and CRITIC combined, the multiplier synthesis normalization method is applied for its ability to reveal the relationship between the weights and their corresponding indices.

$$W_{\text{ANP-CRI}} = \frac{W_{\text{ANP}} \times W_{\text{CRITIC}}}{\sum_{m=1}^k W_{\text{ANP}} \times W_{\text{CRITIC}}} \quad (16)$$

6.3 Construction of URSI Measure Model Based on RSR

The Rank-Sum Ratio (RSR) method is a non-parametric comprehensive evaluation technique that transforms multi-indicator information into ranks and synthesizes them into a single composite index. By using ranks instead of raw indicator values, the RSR method effectively reduces the influence of scale differences and extreme values, making it particularly suitable for multi-criteria assessment problems [10].

In this study, indicator weights derived from the Analytic Network Process (ANP) are incorporated into the RSR framework to construct a weighted comprehensive index, referred to as the Urban Risk–Sustainability Index (URSI).

Step 1. Ranking of indicators

Based on the preprocessed and direction-consistent data z_{ij} , each indicator j is ranked in descending order across all objects i . The resulting rank is denoted by $r_{ij} \in \{1, 2, \dots, m\}$, where a smaller rank indicates better performance.

For tied values, the average ranking method is adopted:

$$r_{ij} = \frac{\text{rank}_{\min} + \text{rank}_{\max}}{2} \quad (17)$$

Step 2. Calculation of weighted RSR

Let w_j denote the ANP-derived weight of indicator j . The weighted rank-sum ratio for object i is calculated as:

$$RSR_i = \frac{\sum_{j=1}^n w_j r_{ij}}{m} \quad (18)$$

Where n is the number of indicators and m is the number of evaluated objects.

Step 3. Ordering and cumulative frequency

The computed RSR_i values are sorted in ascending order. Let k_i denote the rank position of object i in the ordered RSR sequence. The cumulative frequency is then defined as:

$$P_i = \frac{k_i - 0.5}{m} \quad (19)$$

Step 4. Probit transformation

To facilitate grading and regression analysis, the cumulative frequency P_i is transformed using the Probit function:

$$Probit_i = 5 + \Phi^{-1}(P_i) \quad (20)$$

where $\Phi^{-1}(\cdot)$ represents the inverse standard normal distribution function.

Step 5. Regression between RSR and Probit

A linear relationship between RSR_i and $Probit_i$ is established as follows:

$$PRSR_i = a + b \cdot Probit_i \quad (21)$$

where a and b are regression coefficients estimated using least squares.

Step 6. Construction of the comprehensive index (URSI)

Since a smaller RSR_i indicates better overall performance, the Urban Risk–Sustainability Index (URSI) is defined as a normalized positive index:

$$URSI_i = \frac{RSR_{\max} - RSR_i}{RSR_{\max} - RSR_{\min}} \quad (22)$$

where $RSR_{\max}$ and $RSR_{\min}$ denote the maximum and minimum RSR values, respectively.

6.4 URSI Measurement and Analysis

6.4.1 Coupling weight calculation

To ensure that the URSI reflects both expert judgment and objective data variability, indicator weights are determined using a coupled ANP–CRITIC framework. ANP captures the interdependencies and feedback relationships among indicators, while CRITIC assigns weights based on contrast intensity and information content in the data. By integrating these two approaches, the weighting scheme avoids excessive subjectivity or purely data-driven bias. **Figures 5 and 6** jointly illustrate the resulting weight structure and its comparative characteristics, providing a transparent foundation for subsequent URSI construction.

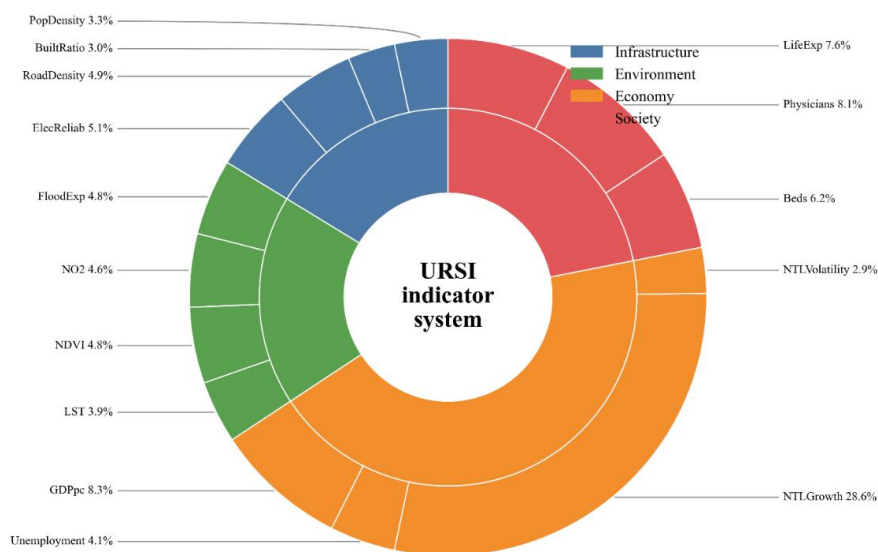


Figure 5 ANP-CRITIC weight

Figure 5 presents the final coupled weights of all URSI indicators derived from the ANP–

CRITIC method. The results show that economic and infrastructure-related indicators receive relatively higher weights, reflecting their dominant influence on urban resilience and sustainability performance. Environmental and social indicators, while assigned smaller individual weights, collectively contribute a substantial share, highlighting their complementary role.

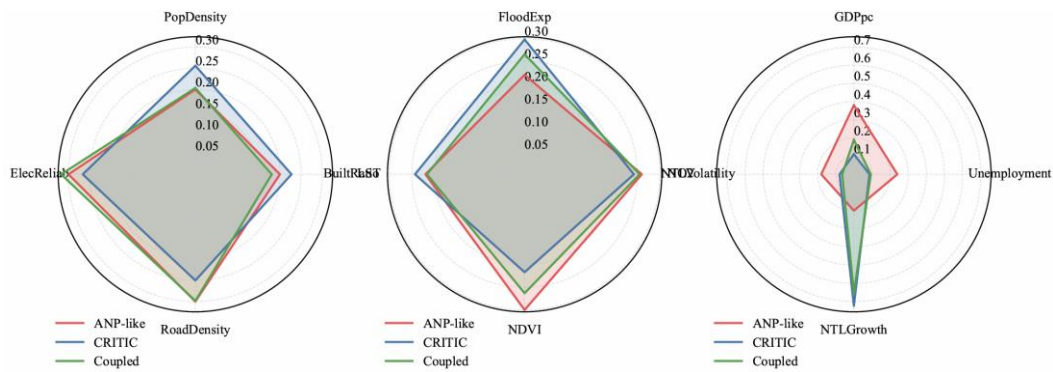


Figure 6 ANP-CRITIC weight comparison

Figure 6 compares the indicator weights obtained from ANP and CRITIC before coupling. ANP-based weights exhibit smoother distributions, emphasizing expert-perceived importance and indicator interdependence, whereas CRITIC-based weights show greater dispersion due to differences in data variability and conflict intensity. The coupled weights lie between these two extremes, indicating that the integration effectively reconciles subjective and objective perspectives.

6.4.2 URSI measure

Based on the established indicator system and coupled weights, the URSI is calculated to provide a comprehensive evaluation of urban sustainability and risk resilience. **Figure 7** summarizes the URSI results for 20 representative cities, offering an overall comparison across different development contexts and geographic regions.

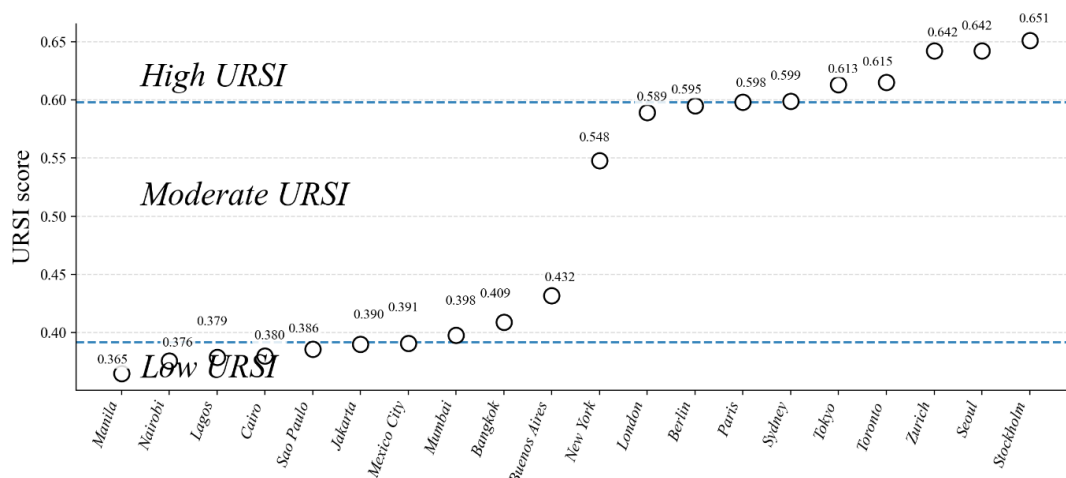


Figure 7 The URSI of 20 Cities

Figure 7 shows the URSI scores of 20 cities ranked from low to high. Cities in developed economies generally exhibit higher URSI values, indicating stronger infrastructure, economic stability, and institutional capacity. In contrast, cities from developing regions tend to cluster at lower URSI levels, reflecting greater exposure to environmental risks and socioeconomic constraints. However, noticeable overlap exists between groups, suggesting that development level alone does not fully determine urban resilience performance.

6.4.3 URSI cluster analysis

To further explore structural similarities among cities, a cluster analysis is conducted based on URSI dimension profiles. **Figure 8** visualizes the clustering results and highlights distinct urban resilience patterns.

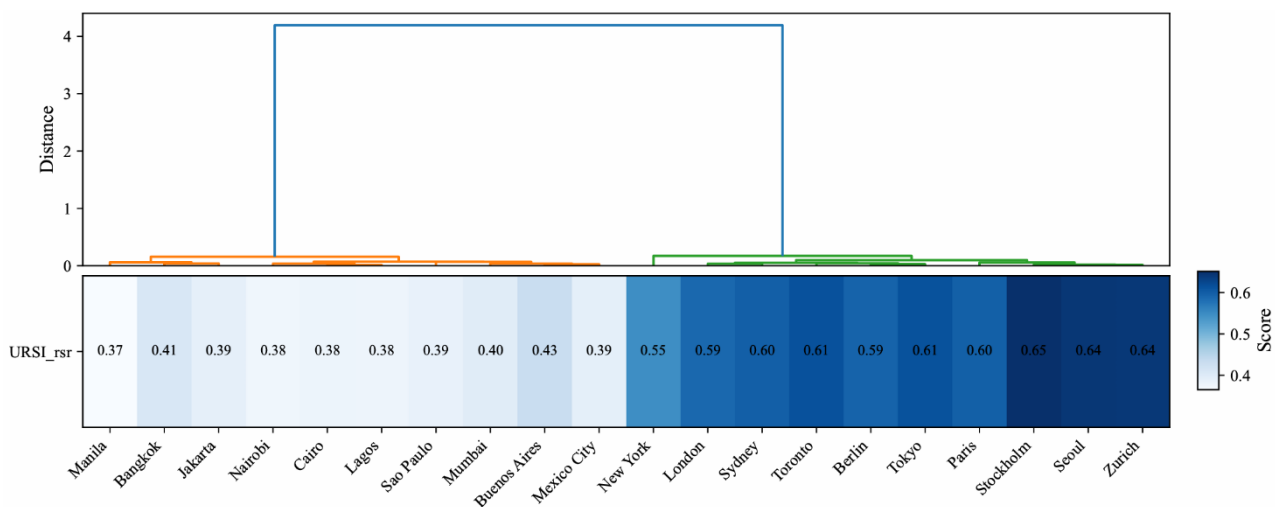


Figure8 Clustering of cities based on URSI dimension profiles

Figure 8 reveals several well-defined city clusters characterized by similar URSI dimension compositions. High-URSI cities form a stable cluster dominated by balanced performance across all dimensions, while low-URSI cities cluster around weaker economic and infrastructure indicators. Intermediate clusters display mixed profiles.

7 LSTM-based URSI Prediction

7.1 LTTM Prediction Model Construction

The Long Short-Term Memory (LSTM) network, introduced by Hochreiter & Schmidhuber (1997), is an extension of recurrent neural networks (RNNs) designed to address the vanishing gradient problem. By incorporating gated memory cells, LSTM can effectively capture long-term temporal dependencies in sequential data. Given that URSI exhibits temporal continuity and dynamic evolution, LSTM is well-suited to model its nonlinear patterns and improve forecast accuracy. Compared to traditional time-series models, LSTM demonstrates superior performance in handling nonlinear relationships, robustness to noise, and multi-step prediction[11].

Step 1: Supervised Sequence Construction

Let y_t denote the URSI value at time t . Using a sliding window of length L , input-output pairs are constructed as:

$$\mathbf{x}_t = [y_{t-L+1}, y_{t-L+2}, \dots, y_t]^\top, \quad \hat{y}_{t+1} = f(\mathbf{x}_t) \quad (23)$$

where $\hat{y}_{t+1}$ is the predicted URSI value for time $t+1$.

Step 2: LSTM Gate Mechanisms

At each time step t , the LSTM unit operates as follows:

Forget gate: decides what information to discard from the cell state

$$\mathbf{f}_t = \sigma(\mathbf{W}_f[\mathbf{h}_{t-1}; \mathbf{x}_t] + \mathbf{b}_f) \quad (24)$$

Input gate: determines what new information to store

$$\mathbf{i}_t = \sigma(\mathbf{W}_i[\mathbf{h}_{t-1}; \mathbf{x}_t] + \mathbf{b}_i) \quad (25)$$

Candidate cell state: creates new candidate values

$$\tilde{\mathbf{c}}_t = \tanh(\mathbf{W}_c[\mathbf{h}_{t-1}; \mathbf{x}_t] + \mathbf{b}_c) \quad (26)$$

Output gate: controls the output based on the cell state

$$\mathbf{o}_t = \sigma(\mathbf{W}_o[\mathbf{h}_{t-1}; \mathbf{x}_t] + \mathbf{b}_o) \quad (27)$$

Here, σ denotes the sigmoid function, $\tanh$ is the hyperbolic tangent activation, $\mathbf{W}$ and $\mathbf{b}$ are learnable parameters, and $\mathbf{h}_{t-1}$ is the previous hidden state.

Step 3: Cell State Update

The cell state $\mathbf{c}$ is updated by combining the previous cell state and the new candidate values:

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t \quad (28)$$

where $\odot$ denotes element-wise multiplication.

Step 4 : Hidden State output

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t) \quad (29)$$

Step 5: URSI prediction

$$\hat{y}\{t+1\} = \mathbf{W}_y \mathbf{h}_t + b_y \quad (30)$$

Step 6: Model training

The model parameters are estimated by minimizing the mean squared error:

$$\min_{\{\Theta\}} \frac{1}{N} \sum_{t=1}^N (y_{t+1} - \hat{y}_{t+1})^2 \quad (31)$$

Step 7: Rolling prediction (optional)

For multi-step forecasting, the predicted URSI values are iteratively fed back into the model:

$$\widehat{y}_{+k} = f([y_{t-L+1}, \dots, y_t, \widehat{y}_{t+1}, \dots, \widehat{y}_{t+k-1}]) \quad (32)$$

In summary, the logic diagram of the LSTM algorithm is shown in **Figure 9**.

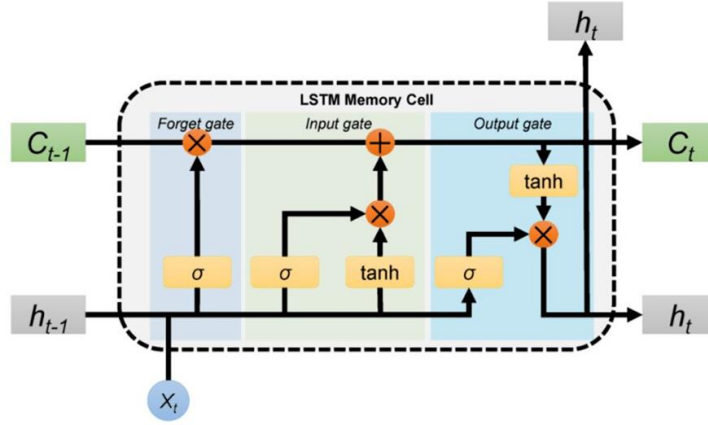


Figure 9 LSTM flowchart

7.2 LSTM-based URSI prediction

To capture the temporal dynamics and nonlinear evolution of URSI, an LSTM-based prediction framework is constructed. **Figure 10** presents the overall performance of the LSTM model, including training accuracy under a hold-out strategy and multi-step forecasting results. By leveraging gated memory mechanisms, the LSTM model learns long-term dependencies embedded in historical URSI trajectories, allowing it to respond to both gradual structural trends and short-term disturbances.

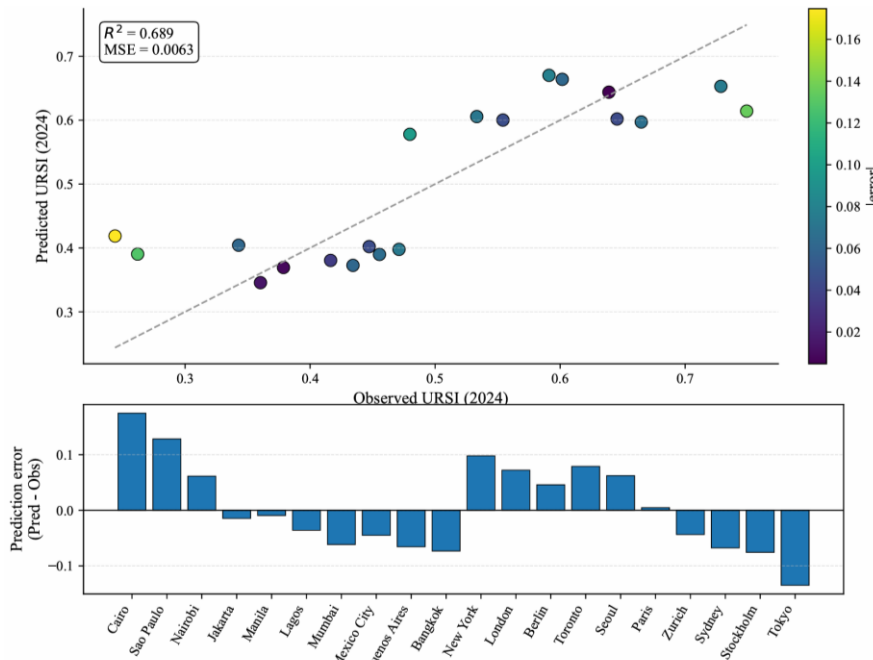


Figure 10 Training Accuracy (Hold-out year: 2024)

Figure 10 evaluates the LSTM model's predictive accuracy using 2024 as a hold-out year. The scatter plot of observed versus predicted URSI values shows a strong linear alignment around the 45-degree reference line, indicating that the model captures the overall magnitude and ranking of URSI across cities.

The reported coefficient of determination ($R^2 = 0.689$) and low mean squared error further confirm the model's satisfactory fit. The accompanying error bar chart reveals that prediction errors are generally bounded within a narrow range, with no systematic bias across high- or low-URSI cities. This suggests that the LSTM model maintains balanced performance across heterogeneous urban contexts.

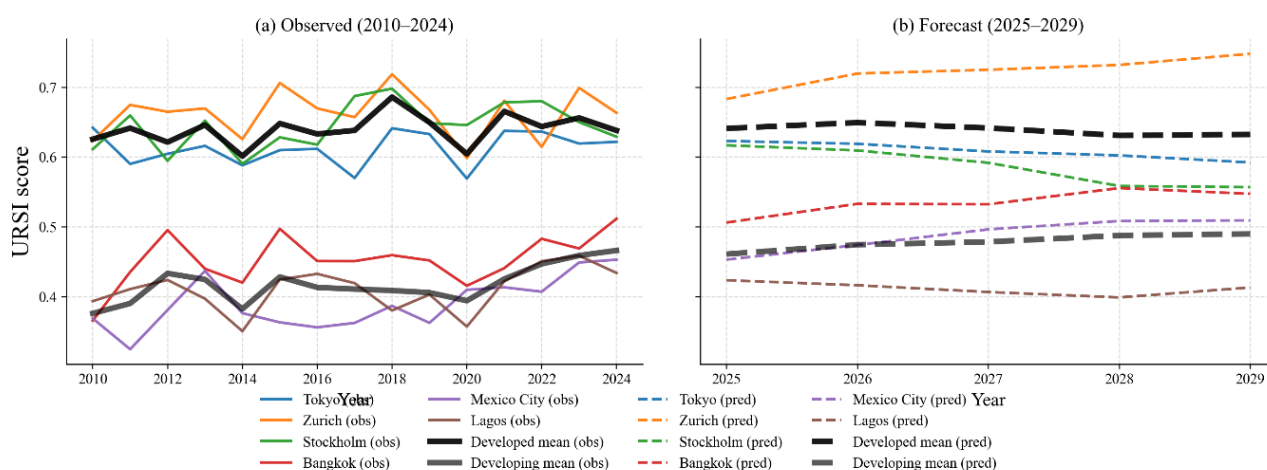


Figure x LSTM Prediction in URSI

Figure 11(a) illustrates the historical URSI trajectories for selected developed and developing cities during 2010–2024. Developed cities exhibit relatively higher and more stable URSI levels, reflecting stronger infrastructure resilience and institutional capacity. In contrast, developing cities show lower average URSI values and greater interannual volatility, especially around global shock periods.

The group mean curves highlight a persistent gap between development categories, while also revealing gradual upward trends in several developing cities. These observed patterns provide essential temporal structure for model learning and serve as a realistic foundation for subsequent forecasting.

Figure 12(b) presents the LSTM-based URSI forecasts for 2025–2029. The predicted trajectories suggest moderate divergence across cities, with developed cities maintaining relatively stable and high URSI levels, while developing cities exhibit slower growth and greater sensitivity to past fluctuations.

Importantly, the forecasted group means remain smooth and continuous, indicating that the model avoids unrealistic oscillations in multi-step prediction. Overall, the results imply that urban sustainability and risk resilience are likely to evolve incrementally rather than abruptly, reinforcing the suitability of LSTM for medium-term URSI forecasting.

8 Model Evaluation and Further Discussion

8.1 Strengths

- **Comprehensive and structured indicator system.** The URSI is constructed from multiple quantitative indicators spanning infrastructure, resources and environment, economy, and social governance.
- **Integrated subjective–objective weighting mechanism.** By combining ANP and CRITIC through a coupling framework, the model incorporates expert judgment while preserving data-driven objectivity. This reduces bias from relying solely on either subjective preferences or statistical dispersion.
- **Robust evaluation and ranking capability.** The RSR-based assessment model enables stable cross-city comparison under heterogeneous indicator scales. It effectively distinguishes resilience levels among cities and supports clear ranking and classification results.
- **Dynamic forecasting of future resilience trends.** The incorporation of the LSTM prediction model allows URSI to be projected over time, capturing nonlinear temporal patterns and providing forward-looking insights for long-term urban resilience planning.

8.2 Weaknesses

- **Limited consideration of extreme disruptive events.** The model assumes relatively stable urban development conditions and does not explicitly incorporate rare but high-impact events such as wars, mega-disasters, or abrupt systemic collapses.
- **Static weighting structure over time.** Although the coupled ANP–CRITIC weights improve reliability, they remain time-invariant. In reality, the relative importance of indicators may evolve as cities face different stages of development and external shocks.

9 Conclusion

This paper proposes a comprehensive Urban Resilience and Sustainability Index (URSI) to evaluate and compare the resilience performance of cities under growing environmental, economic, and public health challenges. An integrated indicator system covering four key dimensions is established, and a hybrid evaluation framework combining ANP, CRITIC, and RSR methods is developed to ensure robustness and comparability. The results demonstrate that cities with stronger economic foundations, environmental capacity, and public health resources tend to exhibit higher resilience and sustainability performance.

Furthermore, an LSTM-based prediction model is constructed to forecast the future evolution of URSI. The predictive results highlight the dynamic nature of urban resilience and suggest that resilience trajectories are shaped by long-term structural conditions rather than short-term fluctuations alone. Overall, the proposed framework provides a practical and extensible tool for assessing urban resilience and sustainability, offering valuable insights for policymakers and urban planners in designing targeted strategies to enhance cities' adaptive capacity and long-term sustainable development.

Cities Under Pressure: A Scorecard for Urban Resilience

Cities today face increasing pressure from climate extremes, public health shocks, and economic instability. Although resilience and sustainability are widely discussed, decision-makers often lack a clear and comparable way to measure them. Without a consistent benchmark, it is difficult to identify vulnerabilities, compare cities, or evaluate whether policies are truly effective. To address this challenge, we propose the **Urban Resilience and Sustainability Index (URSI)**, a concise scorecard that transforms complex data into actionable insights.

URSI integrates publicly available socio-economic statistics with satellite-based environmental indicators to ensure objectivity and comparability. After standardization, the indicators are organized into four intuitive dimensions. Infrastructure and engineering resilience reflects the reliability of essential urban systems. Environmental carrying capacity captures air quality, urban greenness, and urban heat stress. Economic and financial resilience represents income levels, employment stability, and economic vitality. Social governance and public health capacity measures healthcare resources and baseline population health. A balanced weighting scheme combines expert judgment with data-driven information, producing an overall URSI score together with dimension-level results.

Illustrative results show a clear tiered pattern among cities. High-performing cities typically score above **0.75** and benefit from strong healthcare capacity and effective environmental management, enabling smoother recovery from shocks. Mid-ranked cities (**0.67**) often display solid economic momentum but are constrained by environmental pressure or uneven public services. Lower-ranked cities, usually below **0.60**, face multiple challenges simultaneously, such as limited medical surge capacity, higher pollution exposure, and more volatile economic conditions, indicating the need for coordinated policy interventions.

URSI also supports forward-looking analysis. Short-term forecasts suggest that targeted investments can help mid-ranked cities improve faster than top-tier cities, narrowing resilience gaps over time. Measures that reduce pollution and heat exposure, strengthen healthcare systems, and stabilize employment appear especially effective. Overall, URSI provides a transparent and practical tool for benchmarking urban resilience and guiding sustainable development strategies.

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